

Learning content design and learner adaptation for adaptive e-learning environment: a survey

K. R. Premlatha¹ · T. V. Geetha¹

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Abstract This paper presents a survey about learning content designs and various adaptation levels, in order to adapt the learners' necessities in e-learning environment. Normally, learners have different learning styles, cognitive traits, learning goals and varying progress of their learning over period of time, which affects the learner's performance while providing the same bundle of course to all learners. Hence, there is a need to create adaptive e-learning environment to offer appropriate learning content to all individuals. In general, the adaptation can be done based on learners' characteristics. Here, we explore the adaptation that can be done, not only based on learner context parameters but also on the learning content (learning object) and the configuration of e-learning environment. In this paper, we provide a detail review about the various levels of adaptation, learning object design and process for learning content design, learner context parameters, and models/components of e-learning; moreover, we analyze and portray the associations among the components, necessary to achieve the well-defined adaptation in e-learning environment.

Keywords Learning object · Learning content design · Learner adaptation · Learning path · Adaptive e-learning

1 Introduction

E-Learning provides an educational environment where learners are able to access educational materials, anytime and at any place. E-Learning technology has gained significance with the growth and popularity of computer networks through the World wide web (WWW) and the Internet. In E-learning, learners' profile information plays a major role in providing

✉ K. R. Premlatha
premlatha.k.r@gmail.com

T. V. Geetha
tv_g@hotmail.com

¹ Department of Computer Science and Engineering, College of Engineering, Anna University, Chennai, India

educational materials to the learners (Esichaikul et al. 2011). This profile can be used for developing the individual and adaptive learning courses and also take on learners into the learning process so that learner's learning potentials and progress can be optimized.

Adaptive e-learning is a significant requisite to improve the learning performance of the learners. It is expressed as a unique, combined educational model that is customized to the individual learner's needs and goals. Adaptive learning systems should not be restricted by time, place and learner's other requirements. It mostly focuses on learner's preferences and current state of the learner to provide suitable learning content and in addition focuses on the learner's situation. It attempts to adapt the educational materials, to suit the desires of the learners (Das et al. 2010).

Most of the educational materials located in the web environment or learning portals are in the form of digital objects. These digital objects are also known as learning objects (LOs) as they are used specifically for learning purposes in the educational environment (Noor et al. 2011). Learning objects' approach is an innovative one, leveraging the development and deployment of e-learning content in new and interesting ways. There is also a need to consider individuals' learning necessity while designing these learning objects. Therefore based on the individuals' characteristics and needs, the LOs are being adapted while designing the e-learning content.

Adaptive e-learning also provides an alternative to the traditional "one size fits all" approach and has driven the development of teaching and learning towards a dynamic learning environment. Thus, getting an adaptive LO to suit the learners' personalized needs is an important issue. Two major problems arise here: the "one size fit all" approach gives the same learning materials to each learner (Brusilovsky 2001; Stewart et al. 2005), and the immense amount of information available leads to information overload (Berghel 1997). Thus, adaptive e-learning has gained more attention in the last decade (Manouselis and Sampson 2003; Melis et al. 2001; Weber 1999). In e-learning environment, several intelligent proposals have been developed such as intelligent learning contents suggestions (Huang et al. 2008a,b), dynamic learning recommendation (Chen et al. 2005; Huang et al. 2008a,b), adaptive learning (Canales et al. 2007; Wang et al. 2008) and adaptive pedagogical path (Semet et al. 2003).

The primary objective of this study is to recognize various essential processes of learning content design that can deal with the diversified LOs with various information granularities, representation formats, LOs processes using the learning object metadata, so that each learner can utilize the learning object based on the learning scenario, which is adapted to the individual learners' necessities.

2 Overview of learning objects

Learning Objects are small, self-contained and reusable units which represent learning materials. Learning objects play a vital role as resources structuring the learning content and sustaining flexible adaptive tactics for navigation through this content (Boyatcheva and Kovatcheva 2005). The main aim of creating LOs is to attain the intention of maximum reusability and leveraging the maximum cost of the production of quality materials without losing any information of the learning content. The concept of learning object came about from the need to create highly structured e-learning content with pedagogical aspects in a way that can be reused in different learning scenarios. The IEEE Learning Technology Standards Committee (LTSC) describes LOs as "any entity, digital or non-digital, which can be used, re-used or referenced during technology supported learning". Learning object as "any digital resource that can be reused to support learning" (Wiley 2000). To describe these

LOs, many metadata standards are available. The LOs are generally stored and maintained in the learning object repositories. In this section we discuss about learning object metadata standards, repositories and process LOs undergo for content design.

2.1 Learning object metadata and standards

Learning Object Metadata (LOM) is the most common metadata standard to describe educational resources. It is a data model, usually encoded in XML, used to describe a learning object. The purpose of the LOM is to support the reusability of LOs, to aid discoverability, and to facilitate their interoperability, usually in the context of online learning management systems (LMS). There are a large number of standards for metadata information and it can be classified in to four groups: general metadata standards, learning-specific metadata standards, format-specific metadata standards, and subject-specific standards (Sieber and Henrich 2010). In learning specific metadata standards, the IEEE Learning Object Metadata (LOM) standard is the well-known standard, it specifies the syntax and semantics of Learning Object Metadata, defined as the attributes required to fully describing a Learning Object. Learning Object Metadata is a data element for which the name, explanation, size, ordering, value space, and data type are defined in this standard. For example, the LOMv1.0 Base Schema has nine categories: General, Lifecycle, Meta-Metadata, Technical, Educational, Rights, Relation, Annotation and Classification (Barker and Campbell 2010). In e-learning scenario, the other important standards are Dublin Core, IMS, and SCORM (*Sharable Content Object Reference Model*). These standards facilitate interoperability of LOs i.e. the sharing and the exchange of LOs across any technology supported learning system. The global interest in metadata standards and practices has exploded with the growth of e-learning and associated learning object repositories. A LOM repository or learning object repository stores both LOs' and their metadata (Roy et al. 2010).

2.2 Learning objects repositories (LOR)

Learning Object Repository is a digital database which acts as a platform to store, retrieve, gather, edit, publish and maintain sets of LOs. The typical LOR should provide an organized storage of LOs, mutual communication, and speedy search and access to suitable LOs. The Learning Object Repository is an important element in the management, publishing, location and retrieval of LOs. According to standards, the LOR should be able to decompose or assemble a new teaching material. LOR could transfer LOs to users easily and facilitate multiple users to use the same learning object (Liu et al. 2005). However, the task of finding LOs to suit the user's needs and requirements still presents problems related to easiness, simplicity and usefulness in selecting and improving them. Therefore, it is now essential that repositories are developed according to standards and specifications that guarantee interoperability with any other repository or search system. Most LO repositories are generally independent, that is, they work as portals that can be accessed through a Web-based interface and provides a search mechanism with list of categories to conduct the search. Several features of this kind of search can be found in some public repositories such as Agrega, Ariadne, EdNA Online, EducaNext, LORNET, MACE, Merlot and OER Commons (Oton et al. 2012).

2.3 Learning objects for content design

The ultimate aim in the field of learning content design is to develop a common understanding of the Learning material structures, make it interoperable so as to create the learning environ-

ment effective, and compose the educational content made available in any format according to the learners' needs. Hence the concept of "Learning Object" (LO) has been used in the field of learning content design for several years. In this study, we describe the processes of LOs such as LO_design, LO_sequencing, LO_composition, LO_presentation and LO_evolution in to various adaptation levels of e-learning. The process of transformation of traditional learning materials into LOs involves the selection, adaptation and transformation of learning materials so that they incorporate various design models, to meet different learning styles, needs and abilities. In other words, the developed e-learning content which incorporates LOs should comply with the following main principles: Granularity, Interoperability, Accessibility and Adaptability (Wiley 2000) (will be discussed later). Hence, we relate these LO processes to adaptation levels, learner context parameters and models/components of adaptive e-learning. In the following section we describe the overall taxonomy of our survey about the adaptive e-learning environment including learning object processes viewed from various adaptation levels and associated with different e-learning models.

3 Adaptive e-learning environment

Adaptive e-learning courses should be customizable to reflect the learners' interests and needs. In adaptive e-learning, there is also a big challenge in the design of educational materials. In this review, we discuss about an e-learning system that should be adaptable, not only based on the learner's requirements, but also based on the learning content, the goal, and the configuration of the learning environment.

In existing surveys, the general paradigms for implementing adaptive e-learning systems described about the concept of adaptivity, learning style models and components of e-learning (Kamceva and Mitrevski 2012). Moreover, the survey by Graf et al. (2012) discussed how adaptivity and personalization could be achieved based on students' characteristics such as learning styles, cognitive traits, affective states and motivational aspects (Graf et al. 2012). Another survey explained about how adaptive e-learning could be achieved through resource modeling based on domain ontology, learner ontology, instructional role ontology, structural ontology and instructional goal / level by using semantic web technology (Hammami et al. 2012). One more survey discussed about the development of adaptive hypermedia systems based on adaptation models, adaptation levels and adaptation knowledge, ontology for content classification and user description (Souhaib et al. 2010). The above mentioned surveys provided the details about the modes/components, adaptation levels, learner profile information etc. However, they did not provide the details about the relationships between the e-learning models, adaptation levels, learner parameters etc. Furthermore, in our survey we mainly focus on learning object processes at each of the adaptation levels and the associations of these learning object processes with the learner context parameters and e-learning models, since the LOs play a major role in adaptive learning content design. Hence we organize the taxonomy in our survey based on the classification of, the levels of adaptations in e-learning scenario, incorporating the various learning object processes for learning content design and their relationships with learner characteristics, models/components of e-learning. In addition, this survey helps the researchers and practitioners to inculcate the knowledge in the design of e-learning practices, in order to accomplish the adaptation concepts in the different learning environments.

In particular, the adaptation levels are used to tailor a course (or, in some cases, a series of courses) according to the needs of the individual learner. The intention is to optimize the relationship between course contents and user characteristics / requirements, so that the optimal learning result is obtained, while the time and interactions expended on a course

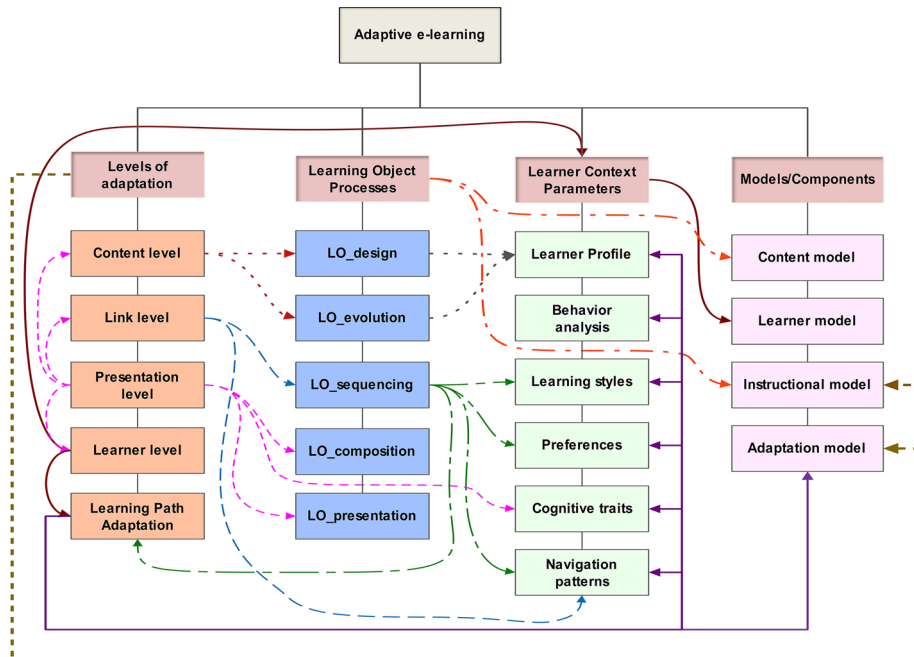


Fig. 1 Adaptive e-learning taxonomy

are brought to a minimum. The levels of adaptation in learning environments are based on a rather well-established set of models (content, learner, instructional and adaptive) and processes. The Content/domain model is usually a representation of the course content being offered. The most important aspect of adaptive-course models is that they are usually based on the identification of relationships between course elements, which are subsequently used to decide upon the learner behaviors adaptation (Brusilovsky 2003). The learner model is used to refer to special cases of user models, tailored for the domain of learning. The specific approach to modeling may vary between adaptive learning environments. The learner model can be updated at interaction time, to incorporate elements or traces of the learner's interaction history. In other words, the learner model in Adaptive Learning Environments (ALE), not only encapsulates general information about the learner (e.g., demographics, previous achievements, etc.), but also maintains a "live" account of the learner's actions within the system (Froschl 2005). The instructional model is responsible for definition of the learner cognitive characteristics and preferences of the Learner Model, the structure of the educational resource description model, as well as, the content selection rules of the Adaptation Model (Karampiperis and Sampson 2005). The adaptation model incorporates the adaptive theory of ALE, at different levels of abstraction. Specifically, the (possibly implicit) adaptation model defines what can be adapted, as well as when and how it is to be adapted. The levels of abstraction at which adaptation may be defined, range from specific programmatic rules that govern run-time behavior, all the way to general specifications of logical relationships between ALE entities that get enforced automatically at run-time. The most successful and widely known ALEs today use adaptation models that generically specify system behavior on the basis of properties of the content model (such as relationships between content entities) (Paramythis and Loidl-Reisinger 2003). Figure 1 shows that overall taxonomy of our

survey about adaptive e-learning. Here we discuss about the levels of adaptation for learning content design through the learning object processes such as LO_design, LO_sequencing, LO_composition, LO_presentation and LO_evolution. In this taxonomy (Fig. 1) we show the associations among the levels of adaptation, learning object processes, learner context parameters and e-learning models. The learning object sequencing will be discussed under link level adaptation; since the associations among the LOs is based on learner context parameters. Subsequently, the learning object design and evolution should maintain the features of reusability and editability. Accordingly, these processes are associated with the content level adaptation and models such as content and instructional models. Moreover, the learning object processes could depend on the learner context parameters. In general, an adaptive e-learning system considers several parameters to define the learner's learning context. Based on these context parameters, an e-learning system offers customized information to the learner. Most of the existing e-learning systems uses only some of the context parameters namely learner's preferences, learning styles, learner's intentions etc.

In this survey we discuss about some additional learner context parameters such as learner personal profile, learner behaviors, cognitive traits and navigational patterns as shown in Fig. 1. Furthermore, the e-learning models/components are used to represent the requirements for designing the system, analyze and design the learning content, manage the presentation of material, ascertain learner mastery by monitoring the learners' knowledge and activities. The levels of adaptation integrate and uses information obtained from the preceding models to drive presentation of adaptive learning content (Shute and Towle 2003). The levels of adaptation such as content level, link level, presentation level, learner level and learning path adaptation will be discussed in the following sections.

4 Levels of adaptation in e-learning

Generally, the adaptive e-learning environment provides a framework to express the functionality of adaptation which specifies what should be adapted, according to what features it should be adapted to, and how it should be adapted. Here we consider two aspects. One is '*what can we adapt to?*'—Several learner characteristics, such as preferences, previous knowledge, goals, interests, background, experiences, learning styles, cognitive traits, context and environment. The next aspect is '*what can be adapted?*'—Presentation (adapting the actual learning content, the presentation of that content, or the media used) as well as the navigation (adapting the link anchors that are shown, the link destinations, and the overviews for orientation support). In general adaptivity has been implemented based on two adaptation technologies: adaptive presentation at the content level and adaptive navigation support at the link level (Brusilovsky 2001; Souhaib et al. 2010). Based on these two classifications, we define five levels of adaptation such as Content level adaptation, Link level adaptation, Presentation level adaptation, Learner level adaptation, and Learning path adaptation. We integrate these adaptation levels in terms of learner context parameters, models of e-learning and LOs processes (as we discussed earlier) such as LO_design, LO_sequencing, LO_composition, LO_presentation and LO_evolution. *Content* level adaptation is the most studied way of hypermedia adaptation (Beaumont 1994; Boyle and Encarnacion 1998). The idea of this level is to adapt the content of a page to the knowledge, goals, and other features of an individual user. With adaptive level of *presentation*, the content of a hypermedia page is individually generated or assembled from pieces for each learner. *Link* level refers to adaptations that take place at the system's interface and are intended to facilitate or support the

user's interaction with the system, without, however, modifying in any way (sequencing) the learning "content" itself. The most typical examples of adaptations in this level are: dynamic course (re-)structuring; adaptive navigation support; and, adaptive selection of alternative (fragments of) course material (Brusilovsky 2001). *Learner level* refers to adapting to the learner context parameters. The idea of *learning path* adaptation techniques is to help the instructional designers to find the learners' paths while learning in e-learning system by adapting the learner preferences, learning styles, and other characteristics of an individual user. The following sections describe about the various adaptation levels, which is used to provide the adaptive environments for e-learning systems.

4.1 Content level adaptation

The quality of an adaptive e-learning course is improved by learner-centered content, granularity, engaging content, interactivity and personalization (Ghirardini 2011). The main requisite while designing the e-learning content is, to adapt the LOs in terms of reusability or editability. In addition, there is a need to consider the LOs life cycle such as obtaining, labeling, offering, selecting, using, and retaining, to adapt the LOs. In content level adaptation, content can be adapted while generating new LOs from existing ones, while deciding the complexity level of contents, and while creating the personalized learning content. This adaptation level can be applied in LOs design and evolution processes, according to the learner personal profile which includes cognitive level and learner preferences (will be discussed later).

4.1.1 Learning objects design

Learning Object Design is one of the main research topics in the e-learning community in recent years, and most of the researchers pay attention to the issue of Learning Object Reusability while designing the adaptive LOs. The main issue in learning objects design is *metadata* generation and *reusability* (Noor et al. 2011). While generating LOs, creation of structured metadata is too difficult, complicated and time consuming for authors of LOs. In general, resource based and context based methods were used for automatic metadata generation (Bauer et al. 2010). The context is defined as, any information that can be used to describe the situation of an entity (Abowd et al. 1999). Normally any systematic approach requires a sound architecture that gives rise to the overall success of creating, using and reusing the LOs (Koohang 2004).

There are two major technical issues to reusing LO: *Editability and Granularity*. *Editability* is important because any aspect of a learning object can be changed if it is available in a suitable form. If a LO is editable, its granularity can be modified. In addition, LO editability allows instructors to contextualize the LO, according to the learners' needs. Before creating the LOs, one must consider the following issues: Granularity, Interoperability and Accessibility (Koohang 2004). *Granularity* (Wiley 1999) refers to how complex a learning object should be. Wiley (2000) discussed two different perspectives for expressing this: Efficiency and Instructional perspective. From the Efficiency perspective, the possible benefits of reusability are attained at the expense of classification. On the other hand, from the Instructional perspective, the major issues are the scope and sequence of the LOs design. Granularity is the process of breaking down the digital learning content in to small pieces or chunks. It deals directly with the size of the LOs. The principle of granularity is to combine LOs to be shared and reused in a diversity of contexts (Currier and Campbell 2005). The authors argue that neither learning time, nor the physical size (in terms of bits and bytes) is a valid criterion

for determining the size and granularity of a learning object. They also suggested that the logical size (in terms of instructional/learning time) rather than physical size (in terms of bits and bytes) is the appropriate manner for defining the size of a learning object. Content models are used for defining the structure, that is, the level of aggregation or granularity of LOs, and include the SCORM Reference Model, the aggregation levels defined by the IEEE LTSC LOM standard (Hodgins and Duval 2002).

Generally, Content model represents the various levels of organization and granularity involved in the creation of digital learning content in both e-learning and knowledge management. This content model also helps to visualize the relationship between granularity and reusability: The more granular the content, the more likely it is that the content will be reused. The more content is contextualized in LOs, components and environments, the less likely it is that it will be reused without modification (Wagner 2002).

Interoperability refers to seamless transportation of LOs to different platforms. A digital tool for designing the learning object has been presented, which helps to shorten and strengthen the reusability and interoperability of learning content (Liu et al. 2005). They developed a SCORM-compliant learning object authoring tool, which constructs the manifest and metadata. It can execute the instructor's need, in order to edit the manifest. In the meantime it can access the remote LOR, to utilize the reusable learning object and interoperable teaching materials, and thus reduced the developmental cost of the system.

A variety of LORs are vastly distributed over the internet, and leads to difficulty in accessing the desired learning object and hindered the concept of interoperability. One of the e-Learning features is to allow the accessibility of various forms of learning object through Learning Object Repository (LOR). LOR provides the essential activities that allow learning object accessibility, use and reuse (Lehman 2007). WBE (Web-based Education) systems are centered in reusability, accessibility, durability and interoperability of didactic materials and environments of virtual education (Noor et al. 2007). In order to easily access and reuse learning object, metadata is used particularly to describe the learning object. The author also determined important metadata for accessibility and reusability of learning object. Thus, while designing the LOs, the adaptation can be achieved by maintaining the granularity, interoperability and accessibility of LOs.

4.1.2 Learning objects evolution

The evolution of LOs is generally based on the similarity of the learning contents. Here, comparing learning objects, as well as producing new objects by combining together existing ones, are two basic activities in a distributed learning system, which exploits the object reusability as the fundamental criterion for the construction of adaptive learning systems.

To determine the similarity among LOs, based on the information represented in their metadata, a metric has been proposed and integrated in to a LOs management system, as part of the searching component (Menendez et al. 2011). This method is based on a fuzzy approach to determine the semantic similarity of XML structures, where these structures are used to represent the LOM. The author states that, the metric has been used to calculate the similarity between Learning Objects based on the IEEE LOM standard. Conversely, the components can be used to compare any XML structure, which is based on the Dublin Core standard metadata. A competency-based approach for comparing and combining the LOs in a distributed system has been presented, based on the dynamic personalization and adaptation of the learning content according to the skills, competencies and knowledge of the learners using Skill Maps and Asset Structures (Albert and Stefanutti 2003). A personalized course generation and evolution scheme has been presented using Genetic Algorithm, which

constructs the personalized courses according to the course difficulty level and learner's changing performance in the learning process (Tan et al. 2012). By comparing the learning objects we can achieve the evolution of learning resources based on content similarity of the learning object. Based on the literature analysis, we explain that the evolution process also incorporates personalization and adaptation techniques, according to the learners' profile characteristics.

While designing LOs, instructional designer need to consider several things by asking questions like "What is a learning object? What are the components of a learning object? What is a learning object made of? What goes into a learning object? And how much of that goes in?" These questions are actually relates to *scope*. Once these questions have been answered satisfactorily, the instructional designer would probably ask questions like "So what do I do with these LOs? What can they be used for? How do I arrange them? In what order should the learner encounter them?" These questions are refers to *sequence* (Wiley 2000). Accordingly, sequencing of LOs process is required to arrange the learning contents based on some tactics, which will be discussed in the following sections.

4.2 Link level adaptation

In e-learning scenarios, content creators are usually required to arrange a set of learning resources (learning objects) and create some links among them, in order to present the organized learning materials to the learners in a comprehensive way. Hence there is a need to apply link level adaptation while creating LOs. This level of adaptation is generally referred to as LO_sequencing which considers pedagogical preferences, navigational patterns, instructional design, and learning path adaptation. Brusilovsky (2001) refers to link level adaptation with navigational link support, which is used to determine the learners' behavior patterns, preferences and learning path (which will be discussed in the following sections such as learner level adaptation, and learning path adaptation). Different LO's have different navigation alternatives, depending on their type, role, content and structure.

4.2.1 Learning objects sequencing

The aim of sequencing learning content is to establish a certain order that will ensure link between the educational objectives and the learning activities of the students. To create an order among the set of LOs is called sequence, so the process is called LO sequencing (de Marcos et al. 2008). Generally the learning content design in adaptive e-learning incorporates, the sequencing of LOs based on the learning necessity, the learning history information, the curriculum details of the LOs, and according to the characteristics of the LOs and learners. Learning objects metadata and the curriculum information of the LOs plays a major role in adaptive sequencing of the LOs. The sequencing is not only based on the content of the LOs, but is also based on the metadata of managing the LOs, which is developed in various representation formats (Seki et al. 2005).

de Marcos et al. (2008) defined relations between LOs; so that the sequencing problem can be characterized as a Constraint Satisfaction Problem (CSP) and Artificial Intelligent techniques can be used to resolve it. The author also discussed the process of creating e-learning contents using reusable LOs and proposed an evolutionary approach for LO Sequencing. Here, the author designed Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) agents that enabled automated curriculum sequencing, by means of interoperable competency records and LO metadata. Curriculum sequencing technology provides the learners with the most suitable individually planned sequence of knowledge units in order to learn and sequence

learning tasks. These methods derived from adaptive hypermedia field (Brusilovsky 1996) and rely on complex conceptual models, usually driven by sequencing rules (De Bra et al. 1999; Karampiperis and Sampson 2006). Generally, adaptation rules (*concept* and *content* selection rules) are required to adaptively select and sequence the learning resources. The problems of inconsistency and insufficiency are considered in the defined rule sets of the Adaptation Model, used for sequencing the learning resources (Karampiperis and Sampson 2005).

The dependencies between educational characteristics of learning materials and learners characteristics are too difficult to understand, since many possible permutations are available (De Bra et al. 2004). The problems of the defined rules are *Inconsistency*-when two or more rules are conflicting, *Confluence*-when two or more rules are equivalent, and *Insufficiency*-when one or more rules required have not been defined (Wu and De Bra 2001). The problems of inconsistency and insufficiency of the defined rule sets are responsible for generating conceptual “holes” in the production learning resource sequence (learning path). This is due to the reason that, though appropriate learning resources exist in the Media Space, the conflict between two or more rules (inconsistency) or the absence of a required rule (insufficiency), prevents the adaptive e-learning systems from selecting and using them in the learning resource sequence. As a result, either less appropriate resources are used from the Media Space, or required concepts are not covered at all by the resulting path (Karampiperis and Sampson 2005).

To overcome this problem, Karampiperis and Sampson (2005) proposed an alternative sequencing method instead of generating the learning path, by inhabiting the concept sequence with available learning resources; it first generates all possible sequences that match with the learning goal in hand and then adaptively selects the desired personalized learning path from the set of available paths. Automated sequencing of LOs is a recurring problem in the e-learning field that could be handled by employing models and sequencing rules along with artificial intelligent techniques, which would ensure the reusability and interoperability of learning objects.

4.3 Presentation level adaptation

In e-learning, content can be presented to the learners in any format as they like to study. Each learner might have different individual characteristics for various learning types. Therefore all the characteristics of the learner are taken into consideration for composing and presenting the LOs. Generally, adaptive composition and presentation of LOs requires some models such as domain model, pedagogy model, user model, adaptation model and instructional model (Vassileva and Deters 1998). While composing LOs, there is a need to consider the levels of adaptation such as content level, link level, learner level and presentation level. In addition, Learner profile information for each learner is the most mandatory need for the presentation of the Learning Material (Li and Huang 2006).

4.3.1 Learning objects composition

The composition of LOs generally begins by defining the instructional needs, then locating and selecting the best learning resources conformed. Finally, it is essential to convert them in to some e-learning standard (LOM), then sequencing and packing those learning resources, in order to present the learners. The composition of an adaptive course requires input from various modeled entities such as the learner, the teacher, the content, concept space/domain ontology, and pedagogical strategy(s). Learning object composition integrates

learning activities of learners, adaptive mechanisms affects the composition process and comprehension of an adaptive courses (Dagger et al. 2005). When composing the courseware content, the instructor may break down the subject matter into varied formats, details and depth to achieve the instructional goals. Supporting this process, LOs represent small and reusable chunks of instructional media (Atif et al. 2003). The author adopted object-based segmentation of knowledge (Bannan-Ritland et al. 2000) in order to provide a constructive approach to e-learning. Thus, the LO is used as the building block, to compose the courseware content, in a process centered on the learner to “free learners from the drudgery of doing exactly similar tasks unadjusted and untailored to their individual needs” (Gibbons 2001) which is the case in traditional classroom instruction.

In general, e-learning curriculum is generated to cater for the different learning needs of the learners such as learning styles, preferences, etc. In addition, the learners may also specify some other constraints, such as difficulty level, learning time, media type, etc. In this situation, the system could provide the facility to the learners to select and organize the learning materials in order to meet their own learning needs. Thus, an approach proposed (Li and Huang 2006) to dynamically compose an adaptive curriculum for e-learning based on a topic-centered resource space. Based on the semantic relationships, which is characterize LOs (LOs) and learner profile, the approach dynamically selects, sequences, and links learning resources into a coherent, individualized curriculum to address learners’ focused learning needs.

For composing the learning content another model has been proposed (Atif et al. 2003) using LO processes. Here, the system consists of three conceptual layers such as authoring layer, LO production layer and LO composition layer. LO production layer is crucial for adapting the LO content to targeted learners. LO composition layer, which is used to adjust the LO content is based on the LO metadata. In addition, they used the learner profiler and learning path generator to cater to learners’ preferences and skills.

The composition of LOs can also be defined as the necessary process for the construction of LOs from a set of digital resources or other LOs, eventually simpler resources such as assets or atoms and that in addition, satisfies certain specific aims of learning for a concrete context (Menendez and Prieto 2008). In addition, the composed LOs should have characteristics such as interoperability, reuse and adaptation, while presenting the learning content to learners.

4.3.2 Learning objects presentation

Learning objects are self-contained instructional units which accommodates heterogeneous learning sources (text, presentation, audio or video) or a combination of any of these media. In general, LO presentation is the way to present individualized learning materials (imbedded into the LOs) based on learners’ requisites and levels of the adaptation such as adaptive learning path, link level and learner level adaptation. Here, the multimedia technology contributes further to learning, when instructional designers use the most effective medium to present specific information. Hence, there is a need for instructional designers to map a learning content to an appropriate media. The content-to-media mapping is carried out through some experiments, so as to provide the best media allocation for learning content (Najjar 1996).

The models or components of e-learning play a major role, while presenting the LOs of any course content. The *content model* expresses domain-related bits of knowledge and skills, and their associated structure or interdependencies. Basically a content model provides the basis for assessment, analysis, instruction, and remediation about the learning content. The *learner model* represents the individual’s knowledge and progress in relation to the knowledge map, and it may include other characteristics of the learner. In addition, it captures the important

aspects of a learner for purposes of individualizing instruction. This includes assessment measures that determine where a learner stands on those aspects (Esichaikul et al. 2011; Froschl 2005). The *instructional model* manages the presentation of material and determines learner comprehension by monitoring the learner model in relation to the content model, and prescribing an optimal learning path for that particular learner. Information in this model provides the basis for deciding how to present content to a given learner and when and how to intervene (Goyal et al. 2012). Finally, the *adaptation model* integrates and uses information obtained from the preceding models to drive presentation of adaptive learning content (Shute and Towle 2003).

While composing and presenting the LO to learners, an instructional designer should consider learner characteristics, adaptation levels, pedagogical preferences, and learning path of all learners. In addition, they need to consider all e-learning models (content, learner, instructional and adaptive) while composing and presenting the LOs. In addition, educational materials could be of any format as the learners' ideas differ from one individual to the other. Therefore all learner characteristics and preferences are taken into consideration for composing and presenting the LOs.

4.4 Learner level adaptation

An adaptive e-learning system considers many parameters that contribute to describing learner's contexts. By using these learner context parameters, the system will give customized learning content to the learner. Context is defined as "any information that can be used to describe the circumstances of an entity." Context sensitive e-learning systems choose or filter the learning materials so as to make the e-learning content more relevant and suitable for the learners in their situation. The selection or filtering of the e-learning materials is done by considering the learner's profile information, learning styles, preferences, learner's situation, etc. Many researchers have worked in the area of context sensitive e-learning systems and all of them considered learner context parameters while designing the learning the content. The customization of learning content is generally carried out by considering various learner context parameters such as learning style, learner's preferences, learner's intention, level of expertise, learner's situation etc (Das et al. 2010).

4.4.1 Learner personal profile

Learner personal profiles are generally used to create instructional/learning content automatically, according to the pedagogical preference and cognitive ability of learners [06]. The profile can be formed based on learner's behavior, learning styles, learner goals, performance etc. Based on learner profile formation the adaptive concept can be defined as Adapted systems, Adaptable systems and Adaptive systems (Edmonds 1981).

- Adapted systems: in which adaptation is hard-wired by the instructional designer; in this case, the system is customized to a particular learner profile, which is defined earlier, at design time.
- Adaptable system: in which adaptation is explicitly required by the learner. More precisely, the learner can specify her/his own preferences, by manually creating her/his profile; thus the system is dealing with a fixed profile, which can only be modified by learners' intervention.
- Adaptive systems: in which adaptation initiative belongs to the system itself, based on continuous observation of learner preferences and needs. The learner's profile is no longer

static; it is dynamically updated by the system, after tracking and analyzing learner behavior (Souhaib et al. 2010).

One of the technical issues in the e-learning system is the handling of learners in a personalized manner, by building different profiles according to their behaviors, knowledge and expertise. Generally, the profile-based frameworks of e-learning, matches the content to the context of interaction, so that the learning content can be adapted to the learner's needs and capabilities (Tzouveli et al. 2008). Moreover, continuous monitoring of behaviors is essential in assessing the progress of learners during the learning process and used to present adaptive learning content which is more suitable to learners. Thus, there is a need to integrate the static and the dynamic learner models in adaptive e-learning systems.

For adaptive e-learning, a personalized semantic retrieval system has been presented that uses domain ontology based matching of static profile information to obtain the LOs from the database (Fouad 2012). However, using typical static profiles for adapting to the needs of the users will not give accurate results. Generally, the learner might not like the same way of approaching learning every time. Therefore each time the learner logs on to the system, he/she needs to be evaluated automatically, and his/her profile needs to be updated dynamically. The learner profile parameters, such as learner behaviors and preferences are identified by using Colored Petri Nets (CPN) (Dharani and Geetha 2013). CPN helps to track the learners' behavior such as navigation, time spent on learning material, preferred learning material, and type of learning. Learning styles and preferences are identified, by using different types of learning contents preferred by the learner. The correct identification of the learning style and dynamic updating of learner profile is a key factor to appropriate learning content presentation to each individual learner in the web based learning system, according to their behaviors such as time spent on particular learning content, preference of learning content etc. In the following section, we discuss about these learner behaviors clearly.

4.4.2 Learner behavior analysis

The behavior of the learner might not always be the same. Therefore learner behavior should be logged every time he/she enters e-learning system. Hence learner behaviors have been observed dynamically based on the various browsing actions like Time spent on materials, Time spent on tests, type of LOs preferred, learning objects skipped, count of various LOs used, Performance in tests, Navigational patterns etc.

The navigational behavior of the students of a e-learning virtual environment are described three levels of behavior analysis of students such as session level, where students perform a few actions in a single session logged to the virtual campus; course level, where all single sessions are joined to form a course navigational pattern; and lifelong learning level, to analyze how students evolve from the beginning of a degree until they successfully finish it each academic semester (Carbó et al. 2005). Based on web browsing semantics, users' behaviors are classified (Bousbia et al. 2010) and adapted navigation typology (Canter et al. 1985). Using the navigation type indicator learners' navigation behavior is evaluated through educational hypermedia resources. This 'indicator with qualitative value classified the learner's navigation behavior in four types (Bousbia et al. 2009) i.e. (i) Overviewing—this value is close to the Canter "scanning" value. It implies that the learner is covering a large proportion of pages constituting the course. Through this fast reading, the user seeks to acquire an overall "panoramic" view of the course. (ii) Studying—corresponds to a partial or complete reading of the course pages, with a span of time on each. (iii) Deepening—is rather close to the

preceding value. It describes a learner who remains a relatively long time on a course, careful with details, and seeking Web documents related to the course topics. (iv) Flitting—close to the Canter “wandering” value. It is a journey without a strategy or a particular goal. The main difference with the overviewing type is the lack of focusing on the course.

Navigational behavior denotes, how learners navigate through the course and in which order they visit different kinds of learning objects and activities. The order in which learners prefer to take in and learn from specific kinds of learning material and activities, as well as in which order and priority these different kinds of learning material and activities should be presented for supporting learners with different learning styles is a key aspect of most learning style theories. In addition, adaptive navigation support, in terms of recommending learners a suitable way through the learning material and activities, is one of the two main ways for adding adaptive functionality to learning systems (Brusilovsky 2001).

Based on the analysis of the learners’ behavior through the collection and the interpretation of traces on the learner’s activities, we can find the learner preferences and learning styles, in a web-based learning. Learner preferences refer what kind of materials the learner wants to study. The learner prefers learning materials, which meet their exact needs in the simplest way possible.

4.4.3 Learner preferences

In general, the enormous and different kinds of learning materials clutter users mind and demotivates them from achieving their learning goals. To overcome such aversion towards the learning materials, learner is provided with the preferred LOs. Here by the kind of material we mean whether the learner prefers concept, detailed concept, examples, flow diagram, case study, exercise etc. Preferences can be identified from the learners’ behavior patterns. Few learners prefer the overview of concepts rather than detailed contents. Others may like to learn facts according to the context while others like to read in a sequential manner. Some might also like to understand the concept using the flow diagram of the entire content while others might like to learn practically like working out examples or doing activities. Moreover, each learner has a preference for a teaching style that allows him to learn better. Some one likes to listen and talk, others prefer to analyze a text, or simply using a visual medium. Therefore, all types of contexts are considered before forming the LOs (El Bachari et al. 2011).

Due to the differences in background knowledge, learning styles and preferences, students may take different approaches towards learning (Watson et al. 2010). The author proposed a pedagogical interface for authoring adaptive e-learning courses to provide a graphical illustration on the pedagogical meanings of different course settings, such as how each part of a course should be delivered to a student without certain academic background, and how much content details are distributed across different parts of the course when a particular student learning preference is applied. The author considered student learning preferences as well as teacher’s teaching preferences.

Though some significant steps have been made towards the direction of providing enhanced, personalized educational services, there is still a great volume of research that needs to be further conducted with a view to rendering these services fully personalized and adaptable. In this context, the learner preferences should play a primary role and should dictate the structure and functionality of the entire educational system. This implies that either the learner is in a position to explicitly specify these preferences or the system has the ability to infer them through a monitoring process. Hence, one of the issues that must be addressed, is the prediction of these learner preferences and the modeling of the learner’s profile, in terms of educational level, familiarity with a particular course, overall performance etc. The

learner model and the learner preferences can be predicted effectively using Bayesian network probabilistic relationships (Kritikou et al. 2008). To learn effectively and manage their way of learning, learners have to be aware of their preferences. This information enables the learners to improve the effectiveness of the learning so as to exploit their own resources and also to find their learning styles.

4.4.4 Learning styles

Adaptive e-learning offers adaptive learning resources, learning policies and adaptive e-courses according to a learner's learning style. The learning style of the learner can be identified by using the behavior pattern of each individual's natural processing capability. Learning style may be defined as "the attitudes and behaviors which determine an individual's preferred way of learning" (Honey and Mumford 1992). The performance of the learners can be enhanced by providing the suitable learning contents to the learners based on their learning styles. Hence, the initial step for creating adaptive learning environments is to identify learners' learning styles. Learning style identification facilitates an instructor to effectively teach by using the personalized learning content (Gregorc and Ward 1977; Keefe 1987; Tseng et al. 2008).

Every description of learning style involves various behavioral features that can be composed and studied from the learning behavior of a learner. There have been some models for describing and measuring learning. Kolb (1999) proposed that learners can be classified into convergent learners, divergent learners, assimilators, and accommodators (Kolb 1999). Based on this classification, these four styles of learning are assessed by two dimensions (abstract/concrete and active/reflective), which describe four abilities required to be an effective learner. A number of researchers have applied Kolb's learning style theory to study the effects of e-learning and hypermedia learning, and most of them indicate that learning styles are a key factor in improving the effectiveness of learning (Kolb et al. 2001).

Learning style can be described as both a student characteristic and an instructional strategy (Keefe 1991). As a student characteristic, learning style is an indicator of how a student learns and likes to learn. As an instructional strategy, it informs the cognition, context and content of learning. Learning style is a consistent way of functioning that reflects the underlying causes of learning behavior (Keefe 1987). Keefe's learning style test, can be used to identify Sequential Processing Skill, Discrimination Skill, Analytic Skill and Spatial Skill of the students, might be the most suitable model to be used in an e-learning or web-based environment, because in such learning context, where rich information sources and various ways of presentations are used, students' processing, Discrimination, Analytic and Spatial Skills are quite important for dealing with the learning materials (Tseng et al. 2008).

Felder & Silverman's model described the category of intuitive/sensitive, global/sequential, visual/verbal, inductive/deductive and active/reflective, which can be used to discriminate 32 learning styles. For example, the sensitive/ sequential/ verbal/ deductive/ active is a learning style (Felder and Silverman 1988). Stangl (2002) distinguished learners into four styles, i.e., acting, hearing, reading and seeing. Instructor feedback and student learning styles significantly affect the perceived learning outcomes of e-learning students (Markovic and Jovanovic 2012). The author also states that, quality of education will significantly be enhanced if instructors modify their teaching styles to accommodate the learning styles of all students in their class. Deborah et al. (2014) identified Felder–Silverman learning style model as the suitable model for E-learning and suggested the use of Fuzzy rules to handle uncertainty in learning style prediction so that it can enhance the performance of the E-learning system. Graf (2013) considers learning styles and introduces a dynamic student modeling approach

that monitors students' behavior over time and builds an accurate student model by frequently refining the information in the student model as well as by responding to changes in students' learning styles overtime using Felder-Silverman learning style model. Learning problems are frequently not related to the difficulty of the subject matter but are associated with the type and level of cognitive process required to learn the material (Keefe 1987). In the next section we discuss about this cognitive ability of the learners, to satisfy the various needs of the learners.

4.4.5 Learner cognitive traits

In general, students have a number of cognitive abilities. Some of these capabilities are crucial for learning. The capabilities such as working memory capacity, inductive reasoning ability, information processing speed, associative learning skills, meta-cognitive skills, observation ability, analysis ability, abstraction ability, etc. (Graf et al. 2012) Adaptation of the E-learning system according to cognitive characteristics of the students is a relatively new direction of research where there is a conjunction of technical and pedagogical aspects (Goyal et al. 2012). It is particularly important that the E-learning systems are able to integrate different types of learning content and navigation in order to be able to respond to diverse needs of the students. Learner Cognitive Traits is used to describe whether the learner is a beginner or whether the learner has some pre-knowledge about a particular topic or the learner is an expert in that topic. The relationship among cognitive processes (remembering, understanding, applying, analyzing, evaluating, and creating) and knowledge outcomes (factual, conceptual, procedural, and meta-cognitive knowledge) has been described (Hench and Whitelock 2010). Accordingly, the author defined knowledge as the acquisition, comprehension, or manipulation of information and the awareness of the cognitive processes used and knowledge level is the measure of the degree to which knowledge has been achieved.

Wenger (1987) proposed a learner model which includes the performance and knowledge of the user which can affect the learning and progress of the students. In some reviews (Carver et al. 1999; Kinshuk 2004) researchers express the opinion that the way of thinking, goals, interests and knowledge levels are the main attributes that determine the definition of different types of learners. It is confirmed that these four types of data are highly correlated with the cognitive style of each learner (Messick 1970). An intelligent tutoring system (ITS) which monitored the performance of a learner, and provided adaptation to learner's learning style, current knowledge level and appropriate teaching strategies in e-learning systems has been presented (Phobun and Vicheanpanya 2010). The system assessed each learner's action within this interactive environment and developed a model of their knowledge, skills and expertise. It composed these three types of knowledge (domain expert's subject matter knowledge, learners' knowledge, and instructional knowledge) and organized into four separate models such as expert model, learner model, instructional model and interface model.

The above sections discussed the four levels of adaptations such as content, link, presentation, and learner level. In addition, we explained about LO processes, learner context parameters, and e-learning models, as well as illustrated the links among them along with adaptation levels. In the next section, we will discuss about the learning path adaptation, issues, various approaches and techniques used for generating learning path.

4.5 Learning path adaptation

As we discussed earlier, one of the new major directions in research on web-based educational systems is the concept of adaptability. Typically the educational system adapts itself

to the learning profile, preferences, styles and ability of the learner. In this section, we look into the issues of providing adaptability with respect to the learning paths of learners. Providing an adaptive learning path according to the context of the learners' is a key issue. An optimal adaptive learning path will help the learners in reducing the cognitive surplus and incomprehension, and thus improving the efficiency of the adaptive e-learning systems.

In reality, the majority of learners are not able to find the most suitable LOs (e.g. learning materials, learning resources) from the web. Consequently, a lot of researchers have focused on developing e-learning systems with tailored learning mechanisms to aid on-line web-based learning and to adaptively provide learning paths. However, most personalized learning mechanism systems ignore the relationship between learner attributes (e.g. learning style, domain knowledge) and LO's attributes. Therefore, it is not easy for a learner to find an adaptive learning object that reflects his own attributes in relationship to learning object attributes. Hence there is a need to create attributes-based system to help the learners find an adaptive LO more effectively [attribute-based ant colony system].

In a web based learning environment the enormous amount of available LOs will increase the cognitive overload for the learner and will lead to disorientation (Van Merriënboer and Ayres 2005). These problems can be overcome using an adaptive learning path based on the learner context (Lin and Wu 2007). Using an optimal learning path the LOs can be provided in an effective way to the learner. That is each learner can be provided with individualized learning contents depending upon their needs and contexts.

There are various approaches found in the literature for learning path adaptation that are classified here based on the techniques used for learning path generation. In Artificial intelligence the usage of *Swarm Intelligence* techniques in adaptive e-learning scenarios offers a way to combine simple interactions of individual learners to resolve more complex problems. Several approaches, for finding the optimal learning path in an adaptive manner, are based on *Evolutionary Computation* (EC) methods. *Genetic algorithm* and *Ant Colony Optimization* are two normally used EC approaches.

The EC methods used for adaptive learning path are categorized as social sequencing and individual sequencing tactics. In the social sequencing, the choice of the optimal learning path is based on the collective path and performance of the complete learners' society. The individual sequencing is based on the individual learner rather than a group's characteristics (Pushpa 2012). *Genetic Algorithm* is the leading technique used in social sequencing is (Romero et al. 2005; Guo and Zhang 2009). Furthermore GA (da Silva Lopes and Fernandes 2009; Chen 2008; Huang et al. 2007) and *Particle Swarm Optimization* (Chu et al. 2011; de Marcos et al. 2007) are also used for individual sequencing. Other EC methods such as *Memetic Algorithm* are mainly used for individual sequencing (Acampora et al. 2011).

In general, the learning scenario is constructed by sequencing the LOs based on the learning necessity, the learning history information, and the curriculum information of the object of learning, according to the characteristics of the LO (Seki et al. 2005). Moreover, the problem of concept continuity of learning paths also needs to be considered while implementing personalized curriculum sequencing because smooth learning paths enhance the linked strength between learning concepts. Generally, inappropriate courseware leads to learner cognitive overload or disorientation during learning processes, thus reducing learning performance (Chen 2008). Therefore, compared to the freely browsing learning mode without any personalized learning path guidance used in most web-based learning systems, the author proposed genetic-based personalized e-learning system, which can generate appropriate learning paths according to the incorrect testing responses of an individual learner in a pre-test, provides benefits in terms of learning performance promotion while learning.

Ant Colony Optimization provides more adaptive and robust solution among all these approaches, in the direction of social sequencing (Pushpa 2012). *Ant Colony Optimization* based inductive planning for recommending learning paths (social sequencing) as it is a heuristic method that has been successfully used to deal with complex problems. Here, the probabilistic rule is biased by pheromone values and heuristic information: the higher the pheromone and the heuristic value associated to an edge, the higher the probability of an ant to choose that particular edge. Once all the ants have completed their tour, the pheromone on the edges is updated. Each edge then receives an amount of additional pheromone proportional to the quality of the solutions to which it belongs. A Learning path is constructed by applying this constructive procedure to each ant (Sengupta et al. 2011).

Learning Path Graph is also used to find out learning path where an acyclic graph defines the structure of domain knowledge and the associated learning objectives (Karampiperis and Sampson 2005). Adapted learning path is selected from this graph and contains all the available learning paths based on learner's attributes in the learner model. Learner model consist of learners' knowledge level and characteristics like learning styles and preferences. The author used suitability function for weighting each connection of the Learning path graph for providing suitability factor for learning resources. From the weighted graph, the most appropriate learning path is selected for a specific learner by using shortest path algorithm. *Learning activity graph* (Zhu and Cao 2008) is used to organize learning resources in a learning task.

By using the *clustering* technique, instead of generating single path for every individual learner, the learners are grouped into clusters (social sequencing) of different learning styles and a learning path corresponding to the cluster can be generated. *Concept Map* is the graphical representation of ontology which reveals the entire courseware structure and core knowledge about a subject domain (Bai and Chen 2008). For grouping courseware with high correlation into the same clusters, *fuzzy clustering* analysis schemes are also used. Moreover, a *Self Organizing Map* (SOM) neural network (Kwasnicka et al. 2008) is used for grouping the learners and providing suitable path fit for the cluster in which the learner belongs.

The statistical approach Bayesian probability theory is used for finding the adaptive learning path (individual sequencing) (Anh et al. 2008). Here firstly a node probability table (prior) based on Bayesian probability theory is created. The probability value is assigned based on the learner level of expertise, learning style and learning pace, which are called candidate learning paths. Next a Bayesian network is constructed to calculate probability value which represents for each knowledge unit in learning path. From this, the shortest path is selected to provide the appropriate learning path for a learner (Marquez et al. 2008).

Petri Net based approaches (Gao et al. 2005; Lin et al. 2004) are also used to create adaptive learning path for each individuals. For adaptive learning path, learner's behavior should also be considered to provide an adaptive environment for learners (Dharani and Geetha 2013). The author used *Colored Petri Nets* (CPN) to track the learners' behavior dynamically in the e-learning system. Finally Adaptive Learning Path is generated using the learner's behavior patterns which are modeled as learner's characteristics like learning styles, goals and performance. Petri Nets based approaches also used for controlling the learning path between learning activities has brought a lot of improvements to the paradigm of adaptive e-Learning systems, based on similarity between LOs graph and the formalism of Petri Nets (Kamceva and Mitrevski 2012).

In this survey we discussed different adaptation levels including LO processes, learner context parameters and e-learning components/models, necessary to create an adaptive e-learning environment for learners by offering appropriate learning content to all individuals.

5 Conclusion

Providing an adaptive and interactive environment tailored to the learner's needs is one of the most important areas of research in e-learning environments. In this survey, we reviewed different aspects of adaptive e-learning environment, along with various LO processes, levels of adaptation and e-learning models. Here, we studied that the adaptation can be provided to an e-learning system, not only based on the learner's requirements, but also based on the learning content (LOs), and the configuration of the learning environment. The existing surveys provided the description about e-learning models, adaptation concepts and learner profile information separately. However, in this survey we described the adaptation levels, LO processes, learner parameters, e-learning models and also portrayed the relationships among them. Hence, this survey offers a clear picture of how to create adaptive e-learning environment, based on the LO processes at each of the adaptation levels. Here, the processes of the LOs such as LO_design, LO_sequencing, LO_composition, LO_presentation and LO_evolution are described, through the various levels of adaptation. Since, the LOs process for learning content design, involves not only the collection, systemization and conversion of learning materials; but also they include various design models, in order to meet different learning styles, desires and capabilities. In addition, the conversion from general curriculum path to individual learning path, which contains the LOs that satisfies the learning objective of the courses, which could be used in various learning environments. Thus, this survey provides detailed information about the LO processes for e-learning content design, which is necessary to offer the appropriate and customized learning content to learners.

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